

TU Dortmund — Linear Models (STAT-412)
Written Examination, Summer 2025

Question 1. Derive the ordinary least squares estimator for β in the linear model $y = X\beta + \epsilon$, and state precisely the assumptions required for it to be unbiased. (12 marks)

Question 2. State and prove the Gauss-Markov theorem. Explain what the word "best" means in the acronym BLUE, and note which assumption is not required. (14 marks)

Question 3. Construct an F-test for the joint significance of two regression coefficients. Give the null distribution and the rejection region at level $\alpha = 0.05$. (10 marks)

Question 4. Define leverage and Cook's distance. Explain how an observation can have high leverage but low influence on the fitted coefficients. (9 marks)

Question 5. Define the link function in a generalized linear model and derive the canonical link for the Poisson distribution. (11 marks)

Question 6. Write your matriculation number on every sheet before handing in your work, and hand in this question paper together with your answers. (0 marks)